

Coordination in Global Teams: A Framework for Exploration

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ABSTRACT

We offer an inclusive framework for studying coordination in global teams, focusing on the management of task dependencies by considering theoretical perspectives from the virtual teams, team cognition, global management, and coordination research literatures.

Coordination processes (mechanistic and organic) are presented as the heart of an input-process-output (IPO) model, preceded by situational inputs (characteristics of the task and work context) and leading to coordination states (effectiveness and cost) and ultimately to performance outcomes.

Keywords: Global teams, virtual teams, team coordination, coordination theory

COORDINATION IN GLOBAL TEAMS: A FRAMEWORK FOR EXPLORATION

Global teams have become increasingly important to multinational firms as a means of organizing work across boundaries of geography, time, and culture; for instance, several authors have noted the prevalence of global teams is in the software industry (Carmel, 1999; Herbsleb & Grinter, 1999; Orlikowski, 2002). They combine the benefits of collocated work teams – skill diversity, flexibility, differential knowledge, and mutual goal orientation – with the advantages of virtual environments, such as cost savings and proximity to clients (Espinosa, Cummings, Wilson, & Pearce, 2003). Global teams achieve success through coordination processes that manage the dependencies associated with individual members and task activities (Malone et al., 1999; Thompson, 1967; VanDeVen, DeBelcq, & Koenig, 1976), consistent with globalization trends and modern telecommunications technologies (DeSanctis & Poole, 1997; O'Hara-Devereaux & Johansen, 1994). This paper examines these coordination processes as part of an input-process-output (IPO) model, bringing together theoretical perspectives from the virtual teams, team cognition, global management, and coordination research literatures into a coherent framework.

While much has been written in separate research streams on coordination, team cognition, and global teams, these theoretical perspectives have not been considered jointly to explain or investigate coordination in global teams. Combining these perspectives through an IPO approach is appropriate for the study of global teams (cf. McGrath, 1991), especially as communication technologies increasingly mediate members' interactions (Kraemer & Pinsonneault, 1990; McGrath & Hollingshead, 1994). These technologies are utilized through the coordination processes – mechanistic and organic practices related to team cognition – in the heart of the IPO model. These processes are preceded by situational inputs, such characteristics of the work context and task, and lead to coordination states (effectiveness and cost). Because we

seek to utilize this coordination model to better support global teams and help them achieve higher levels of performance, we also include performance as a more distal outcome. This basic IPO framework is illustrated in Figure 1.

**** *Place Figure 1 about here* ****

In the following sections we overview coordination theory and apply it to team cognition and global team contexts, outlining both current research and novel propositions. This represents an important contribution to the information systems research literature, as no such theory has been presented prior to this, and no empirical findings have considered each of these perspectives. Our theory builds on extant research findings to provide a more complete view of how global teams coordinate task and situational dependencies across multiple boundaries, providing direction for further empirical investigation to help us understand the role of such teams and the information systems which support them.

COORDINATION THEORY: PROCESSES & MECHANISMS

Coordination has been defined as “the effective management of dependencies among task activities” (Malone & Crowston, 1994). Applied to global teams, coordination includes dependencies among all team resources, tangible (members, finances) and intangible (knowledge), and may be may be pooled, sequential, or reciprocal (cf. Thompson, 1967). Pooled dependency results from the need to use the same resource, as when a decision-making group consults the same set of organizational policies, while sequential dependency may be illustrated through an assembly line, where each successive station relies on the previous, but not vice versa. Finally, reciprocal dependency exists when two or more entities use the same resource nonsuccessively, as when developing software code among a group of programmers.

Coordination has been treated in the literature as both a process (i.e., the act of

coordinating) and a state (i.e., the effectiveness achieved through coordination processes) (Fussell et al., 2004). Accordingly, our IPO model separates coordination processes from its associated outcome, the coordination state. This state can be realized as coordination “success”, wherein task dependencies have been successfully managed. On the other hand, coordination “costs” are the losses incurred in managing task dependencies.

Coordination processes include the set of actions taken by team members to manage dependencies, and may be classified as either mechanistic or more organic (cf. March & Simon, 1958). Mechanistic processes are task-oriented and tend to be fairly stable once implemented, although they may require more preparation. For instance, schedules, plans of work, and other collaboration tools may be used to routinize organizational tasks, such as when structuring a routine task from job analysis data. Some task programming processes are very specific to particular tasks, such as configuration management systems used by software organizations to manage concurrent development (Espinosa et al., 2002b), workflows, error repairs, and outstanding issues (Grinter, 2000).

Prior Finding 1: *Team members coordinate most routine aspects of the task by implementing and using task programming mechanisms.*

Organic processes are used for non-routine aspects of tasks that cannot be easily programmed. They tend to be relatively idiosyncratic and emergent, coordinating dynamic conditions through interpersonal (Van de Ven, DeBelcq, & Koenig, 1976), written or oral (Sproull & Kiesler, 1991), face-to-face, or other communication formats (McGrath & Hollingshead, 1994). For example, organic processes would be useful during an unanticipated hardware breakdown, where events unfold unpredictably and structured task programming would not adjust. In situations like this, teams must communicate to achieve a successful

coordination state. Such communication may consume more time and effort than task-oriented processes, and may divert attention from important tasks, but by their nature, such organic processes are needed to coordinate non-routine demands.

Prior Finding 2: *Team members coordinate less routine aspects of the task through communication systems.*

It should be pointed out that teams may employ a mix of mechanistic and organic coordination processes appropriate for their tasks, and this mix may change over time. Communication processes may vary on a continuum from purely organic to relatively more structured, as when information system software is used to mediate communication in virtual teams (e.g., McGrath & Hollingshead, 1994). It should further be noted that mechanistic and organic processes, in combination, may be especially powerful when applied to collectives. One vivid example of this is an academic conference that organizes thousands of people through mechanistic programming, such as conference program books and websites, yet triggers organic communication through hallway conversations, post-presentation discussions, and the like.

Prior Finding 3: *Teams adapt the mix of task programming and communication mechanisms to coordinate their work, and the effectiveness of these mechanisms: (a) change as the task condition changes and over time; and (b) interact with each other increasing or reducing the effectiveness of other mechanisms.*

Coordination in teams, then, has been shown to occur primarily through structuring the task and communicating among the members. It should be noted that either of these process types can have differential success, such as across national culture boundaries (e.g., Matveev & Nelson, 2004) or be over- or under-utilized in a given task context. For instance, one yet-unpublished study finds that once teams are engaged in productive activity, higher performing

teams communicate less than lower-performing teams during the performance stage (Wholey, Kiesler, & Carley, 1996). In other words, coordination through communication is effective to the extent that the communication is needed and is task-oriented.

Proposition 1: *Volume of communication is not necessarily related to coordination – i.e., poorly coordinated teams may need to communicate frequently to resolve their problems, whereas well coordinated teams may communicate infrequently but effectively.*

TEAM COGNITION & COORDINATION PROCESSES

While teams use task programming and communication processes to coordinate their work, their effectiveness will be limited unless team members understand how to use these task processes and have some common grounding in their communication. For example, new team members must learn a given organization's configuration management system before they can use it effectively with the organization's software. Members in this situation who have had similar training or worked on common projects in the past will be likely to share concepts and terminology, thus improving their ability to communicate effectively. In this way, team cognition may enhance the effectiveness of other coordination mechanisms. At the same time, team cognition can help team members coordinate implicitly because they can make accurate assumptions about what others in the team are likely to do (Wittenbaum & Stasser, 1996). A shared understanding of key aspects of the task, how the team interacts, who knows what, who has done what recently, and how well team members know the task domain and each others' work contexts, can enhance coordination.

The literature on team cognition suggests that effective team members develop organized shared knowledge about the task and the team, which helps them coordinate implicitly (Cannon-Bowers et al., 1993; Klimoski et al., 1994). This shared knowledge helps them anticipate other

members' actions and develop explanations about the task more accurately, helping them plan their own actions more effectively. While the benefits of team cognition have intuitive appeal and widespread research interest, this elusive construct is that has proven to be very difficult to define and operationalize with precision, creating two main problems that have hindered work in this area. First, there is an abundance of constructs and definitions related to team cognition in the literature (Cannon-Bowers et al., 2001), including team mental models (Cannon-Bowers et al., 1993; Klimoski et al., 1994), schema agreement (Rentsch et al., 1994; Rentsch & Klimoski, 2001), transactive memory (Wegner, 1995), and team situation awareness (Endsley, 1995; Wellens, 1993). While there are some overlapping similarities among these constructs, they are all conceptually very different. Consequently, there is a lack of agreement on the conceptual modeling and measurement of team cognition, although some progress has been made recently with proposed measures for team mental models (Cooke et al., 2003; Cooke et al., 2000; Levesque et al., 2001; Mathieu et al., 2000), schema agreement (Rentsch et al., 2001), and transactive memory (Lewis, 2003).

While providing better construct definitions and measures for team cognition mechanisms for coordination is outside the scope of this paper, we suggest a general way of organizing our thinking about team cognition, which is largely based on a very intuitive typology of team knowledge provided by Cooke (2000). In this typology, team knowledge is classified into two main types of knowledge: team mental models and team situation models. Team mental models are organized knowledge shared by team members share about the task and the team (Cannon-Bowers et al., 1993). This shared knowledge develops slowly as team members train together and interact over time and it is more stable and long-lasting. While team members can share knowledge about many things, most of the shared knowledge that matters for coordination

in task-oriented teams are shared knowledge of taskwork – i.e., knowledge necessary to carry out the task; and shared knowledge of team work – i.e., knowledge necessary to interact effectively with team members (Klimoski et al., 1994). Shared knowledge of taskwork is important for coordination because any two team members need to have some overlapping knowledge about each other's task domains in order to understand and better manage the dependencies involved.

Prior Finding 4: *Shared knowledge of taskwork helps team members coordinate important aspects of their respective tasks with each other.*

While shared knowledge of taskwork is important, too much knowledge sharing can be counter productive and even lead to problems like groupthink, cognitive overload and knowledge redundancy. Furthermore, some complex, intellectual tasks may require the pooling of unshared knowledge (Stasser, 1992) from different experts, as the knowledge required may be too complex to be possessed by a single individual. When tasks have interdependent activities that require a high degree of specialization, team members only need to share a small amount of task knowledge to understand how these different specialized tasks need to integrate. In such cases, it becomes very important to have shared knowledge of teamwork. Because individual knowledge is so specialized, knowing what others in the team know – i.e., transactive memory – becomes very important (Wegner, 1995). Faraj and Sproull (2000) extended this concept to that of “expertise coordination”, which not only involves knowing where expertise resides in the team, but also understanding when this expertise needs to be brought to bear to the task, which was found to help performance in software tasks. While transactive memory, expertise coordination and shared knowledge of team work are conceptually distinct constructs, they all share the similarity that having some knowledge about team members helps coordination, as it has been found in empirical studies (Lewis, 2003).

Prior Finding 5: *Shared knowledge of teamwork helps team members coordinate important aspects of their respective tasks with each other, particularly when team members' knowledge is highly specialized.*

While shared knowledge of taskwork and teamwork is important, they are of little help with the most dynamic aspects of the task. For example, while members of a software team can benefit from understanding each others' task domains and from knowing who knows what, these don't help a testing engineer learn when a particular software module is ready for testing. Team situation models are useful for this because they have to do with team members' understanding of the current task situation and how it affects the task and their own activities. This concept originated from earlier work on situation awareness which was defined as "up-to-the minute cognizance required to operate or maintain a [physical or human] system" (Adams, Tenney, & Yew, 1995), involving three processes: (1) perception of what is happening with the elements of the system and its environment at any time; (2) comprehension of what they mean; and (3) projection of their status into the near future (Endsley, 1995).

In the context of working teams, situational awareness has been defined in simpler terms as "an understanding of the activities of others, which provides a context for your own activity" (Dourish & Bly, 1992). When working in teams, individual members not only need individual situation awareness to carry out their task responsibilities, but they also need team situation awareness to work together as a coordinated unit (Wellens, 1993). Team situation awareness is therefore essential when members need to coordinate work with tightly coupled interdependencies. Team situation awareness has received a lot of attention in military and aviation research with synchronous teams working on real-time tasks (e.g., crew flights) (Cooke et al., 2003; Durso et al., 1998; Salas et al., 1995). But despite of the potential benefits of team

situation awareness it has received very little attention in the information systems and global team research literatures. At the same time, there has been substantial effort among collaboration tool researchers to conceptualize electronic tools to provide effective support for certain aspects of team awareness like workspace awareness – i.e., knowing how the workspace has changed (Dourish & Bly, 1992; Gutwin & Greenberg, 1999; Gutwin et al., 1995), task awareness – i.e., knowing who has done when in the task (Espinosa et al., 2000), presence awareness – i.e., knowing who is around (Boyer et al., 1998; Handel & Herbsleb, 2002).

Other tool design researchers have defined several other types of team awareness such as activity awareness (knowing the actions of other team members at any point in time); availability awareness (knowing if other team members are available for interaction); process awareness (knowing which steps need to be taken to move the team process along); perspective awareness (knowing how to make sense of others' actions); and environmental awareness (knowing about events external to the immediate workspace that have implications for the team's activity) (Steinfeld, Yang, & Pfaff, 1999).

Other researchers have classified awareness, not based on what type of awareness information is involved, but on other attributes. For example, following a collective intelligence model proposed by Smith (1994), Chen and Gaines (1997) formulated a framework that includes two forms of awareness. The first type is collective awareness – i.e., awareness of the group's collective, long-term memory, which can exist in teams at three levels of detail: deep awareness (i.e., a highly detailed level of information), peripheral awareness (i.e., a less detailed, but still substantial level of information), and global awareness (i.e., a relatively low level of detail). The second type is awareness of other, which can be of three types: resource awareness – i.e., who has what expertise in the team, similar to transactive memory; task-socio awareness – i.e.,

information about social/political dynamics among teammates; and chronological awareness – i.e., the instantaneous awareness that an individual has regarding the activities of others and knowing that something has changed, when and by whom.

Gutwin and Greenberg (2004) conclude from their several studies on the subject that awareness is important for coordination in collaborative tasks that contain interdependent activities because: it helps individuals shift from individual to shared activities seamlessly and with ease; it also helps them anticipate how the task will progress and which actions other team members will undertake; and it helps them make things happen in the right order and at the right time without conflicting with others in the team because members have a better understanding of the sequence of things and the temporal boundaries of their actions. Because of the widely diverse taxonomy of awareness types discussed in the literature, and because we are interested in coordination, it is not particularly useful to focus on specific awareness types, but on more general types of taskwork and teamwork dependencies that team awareness can address. Chen and Gaines (1997) argued that chronological awareness improves team member connectedness, helping synchronize task activities, thus positively affecting coordination.

Prior Finding 6: *Awareness of taskwork – i.e., activities undertaken to carry out the task – helps team members coordinate task activities.*

Prior Finding 7: *Awareness of teamwork – i.e., activities undertaken to interact with each other – helps team members coordinate team member actions.*

INTERACTIONS: MECHANISMS SUBSTITUTE FOR EACH OTHER

From our discussion, it is evident that team members have an array of coordination mechanisms available to manage the dependencies among task activities. This selection of coordination mechanisms includes traditional mechanisms – i.e., task programming and communication – and cognitive mechanisms – i.e., shared knowledge and team awareness. The

effectiveness of each individual mechanism in helping teams coordinate will depend on the nature of the task (McGrath, 1991), and more specifically on the nature of the dependencies present among its various activities (Espinosa et al., 2004). For example: co-located software developers are very effective at coordinating their activities through intensive communication (Teasley et al., 2002) and through informal communication in the hallways and at the water cooler (Kraut et al., 1995; Perry et al., 1994); distributed software developers rely heavily on software collaboration tools like configuration management systems (Espinosa et al., 2002b; Grinter, 2000); other large-scale software teams use transactive memory to coordinate the location and use of others' expertise (Faraj et al., 2000) and yet other software teams rely on collective mind (Crowston & Kammerer, 1998) – i.e., a shared understanding of the group's task and each other (Weick & Roberts, 1993); other information system teams (Nelson & Coopridge, 1996) and management teams (Espinosa et al., 2002a) rely on shared knowledge; and real-time teams like pilot crews rely on situation awareness (Cooke et al., 2003).

Proposition 2: *Teams employ a mix of traditional and cognitive coordination processes to address the coordination needs of the task, depending on personal preferences, prior experience and the nature of the task dependencies involved.*

Proposition 3: *Teams can achieve similar levels of coordination using a different mix of coordination mechanisms – e.g., three teams can be highly coordinated, one primarily through intensive and effective communications, another on through sound task programming mechanisms and the other one through strong team cognition, or through some combination of the three.*

This last proposition implies that when certain coordination processes are employed, others may not be as necessary. For example, a team with well developed plans and routines may not need to communicate frequently. Similarly, teams with very effective collaboration tools designed to exchange task information, may not need to develop as much shared knowledge,

compared to teams that do not have access to such tools (Espinosa et al., 2002b). The reduced need for a particular coordination process may be due to the fact that another coordination mechanism takes care of most coordination needs, or because the use of one particular mechanism makes another more effective – i.e., an interaction effect. For example, the term “mutual knowledge” has been used to indicate that team members not only share knowledge, but are aware that they do so. Mutual knowledge provides common ground among team members, increasing their communication effectiveness as they more use common vocabularies and share meanings (Cramton, 2001; Krauss & Fussell, 1990). Similarly, team members with strong, well organized shared knowledge don’t need to communicate as much to coordinate (Espinosa et al., 2002a; Mathieu et al., 2000).

Proposition 4: *Coordination mechanisms interact with each other in their effectiveness on helping teams coordinate. Depending on the nature of the task and its dependencies, some coordination mechanisms will enhance or offset the effectiveness of other mechanisms.*

TASK DEPENDENCIES:

LINKING COORDINATION EFFECTIVENESS AND PERFORMANCE

While being highly coordinated is an important goal in teams, coordination is only good to the extent that it leads to team performance. Although team performance is not the focus of this paper, it is important to view the framework from a performance perspective. Throughout our discussion we have emphasized the fact that the task really matters when it comes to which coordination mechanisms are most effective. Consistent with coordination theory, we argue that coordination has an effect on team performance when the task activities are highly interdependent. Tasks with tightly coupled dependencies will require more coordination than

tasks with relatively independent sub-tasks (Thompson, 1967; VanDeVen et al., 1976), which has been observed in software tasks (Crowston et al., 1998; Curtis et al., 1988; Kraut et al., 1995). In fact, if task activities are highly interdependent, individual performance does not necessarily lead to high team performance unless individual members' activities are well coordinated.

For example, a team of competent software developers may produce individual software code effectively (e.g., error free, on time, good quality) but the final integrated product may not be ready on time because their individual schedules were not properly synchronized, or the multiple individual software parts may not work well together because the software interface requirements among these parts was not clearly specified (Espinosa et al., 2002b). Conversely, while coordination may be necessary for performance in some tasks, it may not always be sufficient because there may be other factors that may hinder or enhance performance. For example, a well-coordinated software effort may yield high quality, error-free software in a timely manner, but it can be rendered useless if the market doesn't see any value in the new functionality that this product offers. Conversely, if activities can be carried out independently, then coordination is not likely to make a substantial difference on performance (Malone & Crowston, 1994).

Prior Finding 8: Coordination has a substantial effect on team performance when task activities have tightly coupled dependencies and, conversely, coordination has a negligible effect on team performance when task activities can be carried out independently – i.e., task dependencies moderate the effect of coordination on performance.

Prior Finding 9: Coordination is necessary for performance in interdependent tasks, but

it is not sufficient – other antecedent variables will also affect team performance.

GLOBAL TEAMS AND TEAM BOUNDARIES

Much of our framework and propositions developed so far apply to collocated teams as well as virtual or global teams. Next, because our goal is to increase understanding of global team coordination and provide avenues for empirical work, we will focus on factors that differentiate global teams. While much has been written about virtual teams (e.g., DeSanctis & Monge, 1999; Dube & Pare, 2001; Jarvenpaa & Leidner, 1999; Klein & Kleinmanns, 2003; Markus et al., 2000; Maznevski & Chudoba, 2000; Montoya-Weiss et al., 2001; O'Leary, 2001; Powell et al., 2004; Watson-Manheim et al., 2002a), we use the term “global teams” to specify our interest in teams that coordinate across distance and culture. “Virtuality” in teams is still being defined in the literature (Griffith et al., 2003; Kirkman & Mathieu, 2005; Lu et al., 2003), whereas the global team construct is more precisely conceptualized.

DeSanctis and Monge (1999) characterized virtual organizations as a collection of geographically distributed collaborators with diverse backgrounds linked by electronic forms of communication, who rely on lateral and dynamic relations for coordination. Jarvenpaa and Leidner (1999) characterized virtual teams as teams that: have permeable interfaces; form, reorganize and dissolve rapidly as the task needs change; involve team members with different competencies, who are separated by time, distance and cultures; and are mediated by electronic communication technologies. These characterizations are very useful for many types of modern teams found in organizations today. At the same time, these definitions imply two things. First, virtual teams are temporary and they dissolve when no longer needed for the task. Second, that some virtual teams are geographically dispersed, but some are also domestic or non-global. Because the focus of this paper is on global teams, we build on the above characterizations, but

remove these two constraints and define global teams as teams whose members are located in more than one global location, who interact primarily through electronic communication media, who are separated by time, distance, cultures and other team boundaries (Espinosa et al., 2003).

This definition is very useful because it gets at the heart of the coordination problems faced by these teams; their members need to bridge multiple internal boundaries to manage the dependencies among their task activities. Boundaries are the “often imaginary lines that mark the edge or limit of something” (Cambridge Dictionaries On-Line) and can be thought of as the lines within (internal edges) and around (external edges) the team. Watson-Manheim and colleagues (2002a) call these boundaries “discontinuities,” and in a related review of 75 studies on the subject (2002b) they documented the difficulties of getting the work done across these discontinuities. Similarly, in a recent study of a global software development organization, Orlikowski (2002) found that team members had to bridge multiple boundaries to do their jobs, including 15 global locations, 19 time zones and various levels and areas of technical expertise, which presents a substantial coordination challenge.

We argue that the presence of these boundaries creates two main problems that affect team coordination. The first problem is that some coordination mechanisms available to co-located team members are not generally available to global team members. For example, some global team members may have never worked face-to-face and may have never met each other, thus making it difficult to develop mutual knowledge. Similarly, certain types of awareness like task awareness and presence awareness are not readily available to global teams. Also, global team members cannot meet as frequently and as spontaneously as co-located members. The second problem is that the effectiveness of some of the available coordination mechanisms may be affected by these boundaries. For example, communication via electronic mail can be affected

if there are language boundaries in a team. Similarly, the effectiveness of electronic mail can be severely impaired if there are large time-zone differences among team members.

Proposition 5: *Global team boundaries reduce the availability of coordination mechanisms.*

Proposition 6: *Global team boundaries affect the effectiveness of different coordination mechanisms in different ways, making them generally less effective.*

Proposition 7: *Global teams will select a mix of coordination mechanisms that they deem most effective in bridging the particular boundaries present in their teams.*

We now discuss how different coordination mechanisms are affected by team boundaries. Coordinating work across the typical boundaries found in global teams is difficult. For example, a study of global IT project managers found that teams can be very successful when working globally, but that this success requires considerable additional time and effort to coordinate things (e.g., working longer hours, frequent traveling, over-detailed specifications and documents) when working across team boundaries (DeLone et al., 2005). The more boundaries present, the deeper the “fault lines” that team members need to bridge to get the work done (Lau & Murnighan, 1998). Generally speaking, team boundaries associated with global teams make it difficult for members to communicate effectively. Studies have found that team boundaries lead to misunderstandings (Armstrong & Cole, 2002), reduced mutual knowledge and common grounding in their communication (Cramton, 2001; Olson & Olson, 2000) and reduced opportunities for informal communication (Kiesler & Cummings, 2002).

Prior Finding 10: *Global team boundaries have a negative effect on the team’s ability to coordinate through communication.*

Task programming mechanisms like plans, schedules, tools and division of labor take

some time to implement, but once they are in place they are very effective at helping teams coordinate. Because task programming mechanisms help coordinate the most routine aspects of the task, once the use of these mechanisms are learned and internalized, their effectiveness are less likely to be affected by team boundaries. In addition, most task programming mechanisms are designed to be used asynchronously, thus their capabilities are not affected much by boundaries. In fact, task programming mechanisms can be quite effective at helping global teams with multiple boundaries coordinate. For example, studies have shown that software tools like configuration management systems, which developers can access at their convenience, are very effective coordination tools for global software developers (Espinosa et al., 2002b; Grinter, 2000). Similarly, other studies of large-scale global software development have found that division of labor that maximizes dependencies within each site and minimizes dependencies across sites (e.g., keeping all the design in one site and the coding in another site, keeping all the development of a given module in one site) is an effective way to become coordinated because this decoupling of dependencies reduces the need to manage dependencies across sites (Herbsleb et al., 1999).

Proposition 8: *Global team boundaries do not have a substantial negative impact on the team's ability to coordinate through task programming mechanisms – conversely, these mechanisms can be quite effective at helping global teams coordinate.*

Shared knowledge can be most effective in helping global teams coordinate because it can make its members' limited communication more effective. Given that communication is impaired by the presence of boundaries, shared knowledge of taskwork and teamwork become very important in creating common grounding and making this limited communication more effective (Cramton, 2001; Krauss et al., 1990). At the same time, shared knowledge is very

difficult to develop across team boundaries because of the reduced communication quality and fewer opportunities available for interaction. One popular tactic discussed by global IT project managers to reduce the “perceived” geographic distance is to travel frequently to other sites to get to know their teammates and their work environments better (DeLone et al., 2005). When team members become familiar with their remote colleagues’ work contexts, cultures, practices it makes it easier for them to collaborate. For example, a study of global software teams found that shared knowledge helps all teams coordinate, but that when this shared knowledge is available, its effect is stronger on geographically dispersed teams than on co-located teams (Espinosa et al., 2002b).

Proposition 9: *Team boundaries make it more difficult for team members to develop shared knowledge of taskwork and teamwork, but when this shared knowledge is available among team members it has a strong impact on coordination.*

Team awareness is perhaps the most noticeable aspect of team cognition that is readily available in co-located environments, but less available in asynchronous collaboration, and largely unavailable in geographically dispersed collaboration, presenting a substantial challenge for coordination. Thus, we argue that team awareness of taskwork and teamwork can have dramatic effects in geographically dispersed collaboration. This is perhaps the reason why so many collaboration tool developers have rushed to build tools with awareness features and why so many collaborative work researchers are studying awareness. Simple things we take for granted in co-located environments like awareness of who is around and awareness of working hours and holidays can make substantial differences in distributed environments as some researchers have found (Boyer et al., 1998; Handel et al., 2002). While most awareness tools have aimed at helping geographically dispersed teams collaborate (Greenberg et al., 1996;

Gutwin et al., 1999; Gutwin et al., 2004; Gutwin et al., 1995), team awareness tools can also be designed and implemented to bridge other boundaries in global teams. For example, prototype tools created in research labs have been very useful in helping global teams learn about holidays, time-zones, and work schedules in other sites (Colbert et al., 2001).

Proposition 10: *Team boundaries make it difficult for team members to develop team awareness of taskwork and teamwork, but when this awareness is available it has a strong impact on coordination.*

While we have formulated general propositions that affect most team boundaries in general, distance and time boundaries have received considerable attention in the literature (Allen, 1977; Bullen & Bennett, 1993; Conrath, 1973; Van den Bulte & Moenaert, 1998) and they deserve special attention here also. In a nutshell, we formulated propositions arguing that team boundaries make some coordination mechanisms unavailable and some others less effective. In addition, distance and time boundaries force team members to interact through electronic media, thus affecting further their ability to coordinate effectively (Kiesler et al., 2002; McGrath & Hollingshead, 1994; Sproull et al., 1991). For example, geographic dispersion in software teams has been associated with increased coordination overhead, more substantial delays, failure to communicate and retain contextual information, and differences in feedback cycles (Carmel, 1999; Espinosa et al., 2002b; Herbsleb et al., 2003). Furthermore, electronic mediation changes the richness of the interaction among team members (Daft & Lengel, 1986; Dennis & Kinney, 1998), thus affecting the way they manage their mutual dependencies. At the same time, researchers have argued that the pure effects of geographic distance are often confounded with other team boundary variables (Armstrong et al., 2002; Cramton, 2001; Espinosa et al., 2003) and that the true effects of pure distance can be effectively addressed with

information technologies (Olson et al., 2000). For example, the lack of team awareness can be addressed with awareness tools. Similarly, the reduced opportunities for interaction can be addressed with videoconferencing and imaging technologies installed in public areas of various locations (Dourish & Bly, 1992).

Proposition 11: *The effects of pure geographic dispersion are generally confounded with the effects of other global team boundary variables*

Proposition 12: *The negative effects of pure geographic dispersion can be offset to a large extent with information and communication technologies.*

Many of the negative effects often associated with geographic dispersion don't necessarily have to do with distance per se, but with the fact that geographically dispersed teams interact more asynchronously than co-located teams (Espinosa et al., 2002a). This could be easily studied by comparing geographically distributed teams to co-located but highly mobile teams whose members interact mostly asynchronously. While time boundaries also force team members to interact through electronic media, time separation brings about the added challenge of having to pay close attention to timing issues (Espinosa & Carmel, 2004). To begin with, time separation is not only about time zones, but also about differences in holidays, weekdays, work schedules and vacation schedules, among other things. For example, one study documented the case of a global team in which members were only available to meet synchronously 50 days in the year because of these differences, and during these 50 days team members still had differences in time zones, reducing the window of available meeting time to only a few hours per year (Espinosa & Carmel, 2004).

At the same time, time differences can be beneficial. Time-zones have been used effectively to develop programs or provide technical support 24 hours a day in what has been

called “follow-the-sun” work (Carmel, 1999). However, there are very few documented sustained successful experiences of this collaboration mode for highly interdependent tasks. It has been argued that the reason for this is that time separation reduces delay costs (i.e., one site can work when the other one sleeps – the benefit), but it also increases communication and delay costs when miscommunication needs to be repaired (i.e., one site has to wait until the other one wakes up to communicate – the cost). So, tasks in which the timing of activities can be programmed with some certainty (e.g., coding of small programs) or those that are not too sensitive to timing issues (e.g., support centers) have substantial benefits in reduced delay costs, while tasks that are more equivocal and uncertain, which require substantial interaction have substantial costs (Espinosa & Carmel, 2004). In addition, time separation is affected by the amount of work time overlap between sites. While small time differences can have substantial impacts¹, as the work time overlap is reduced, the coordination challenges grow rapidly because there are more non-overlapping hours in which team members have to decide between communicating asynchronously and through less rich media (e.g., electronic mail) or wait until the next overlapping period to communicate synchronously.

Proposition 13: *Time separation – caused by differences in time zones, holidays, and work schedules – requires that global team members pay close attention to the timing of activities to coordinate.*

Proposition 14: *Time separation is beneficial when tasks are fairly routine, certain, the timing of their activities are well organized and teams can effectively use task programming mechanisms to manage key task dependencies.*

¹ A field study of software teams documented a case in which a one hour time difference between two countries reduced the overlapping working hours by more than 4 hours, 2 hours at the beginning and end of the day, 2 hours during lunch time, and additional time because some team members in the earlier time zone often arrived early to work and left early (Herbsleb & Grinter, 1999)..

Proposition 15: *Time separation is detrimental when tasks are non-routine, uncertain and contain tightly coupled dependencies, which require frequent interactive communication to coordinate.*

Proposition 16: *The positive or negative effects of time separation on coordination are amplified as the work time overlap between sites is reduced.*

RESEARCH FRAMEWORK FOR THE STUDY OF GLOBAL TEAM COORDINATION

Based on the arguments and propositions formulated, we suggest the research framework illustrated in Figure 2 for the study of coordination in global teams.

**** Place Figure 2 about here ****

CONCLUDING REMARKS AND LIMITATIONS

Any model will suffer limitations of inclusion. Perhaps most egregious in this case is our decision to forego explicit incorporation of the temporal aspects of global teams. More so than the typical virtual teams, global teams often operate for a year or more, making temporal aspects important considerations. Coordination needs may change as teams experience dynamic life cycles and their tasks evolve over time. For instance, teams may communicate frequently early in their history, divide labor towards the middle, employ task directive processes as they advance, and rely primarily on shared knowledge as they move toward task completion (Wholey et al., 1996). Further exploration into temporal aspects will likely require additions to our framework and propositions.

Despite this limitation, our framework contributes by integrating multiple perspectives that influence global team coordination and performance. Substantial research efforts are currently directed toward global teams, team cognition, communication systems, and coordination, but there has not previously existed a framework sufficient for interpreting the

various streams. This paper seeks to fill this need, specifying propositions within a broad theoretical frame. We believe this to be particularly important as information technologies continually improve the capacity for organizing work across boundaries but do not confidently specify how such technologies can effectively support collaboration and performance.

One important feature of this framework is its foundation in coordination theories; all propositions are derived from the perspective of understanding and managing dependencies among task activities, as suggested by the organizational (i.e., task programming and communication) and team cognition (i.e., team shared knowledge and team awareness) research literatures. This focus on coordination engenders based on simple principles – that teams can achieve performance through managing dependencies and individual performance across the boundaries that exist between global team members. The understanding gained can help to develop more effective information system tools and promote more effective team relationships and, ultimately, global team performance.

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Figure 1: Basic Research Framework for the Study of Coordination in Global Teams

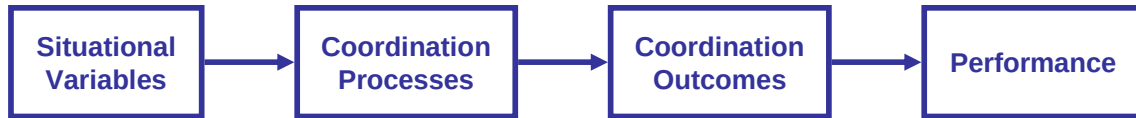


Figure 2: Detailed Research Framework for the Study of Coordination in Global Teams

